Towards Actionable Surgical Team Dynamics: from Teamwork to Counterfactual Annotations

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Section organization

1. Operating Room 2.0
2. Datasets
3. Modeling teams
4. Feedback
5. Future works

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1. Operating Room 2.0 - Motivation

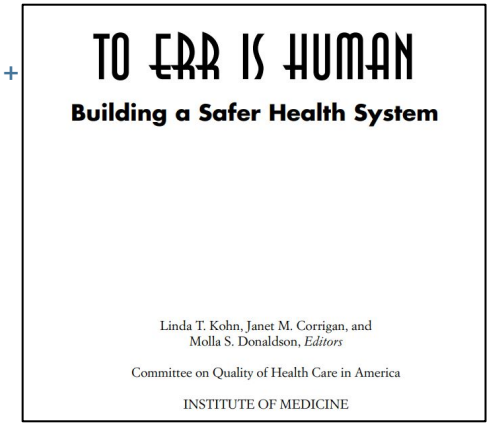


Figure: (Brown, 2001)

- Medical errors are the **third leading cause of death** in the US
- One of the main reasons for medical errors is **poor teamwork in the OR**

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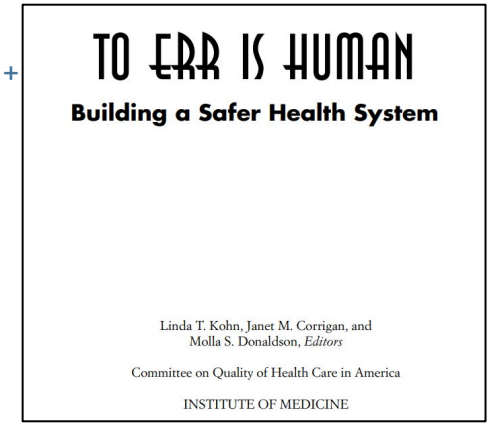


Figure: (Brown, 2001)

+ Medical errors are the third leading cause of death in the US
+ One of the main reasons for medical errors is poor teamwork in the OR

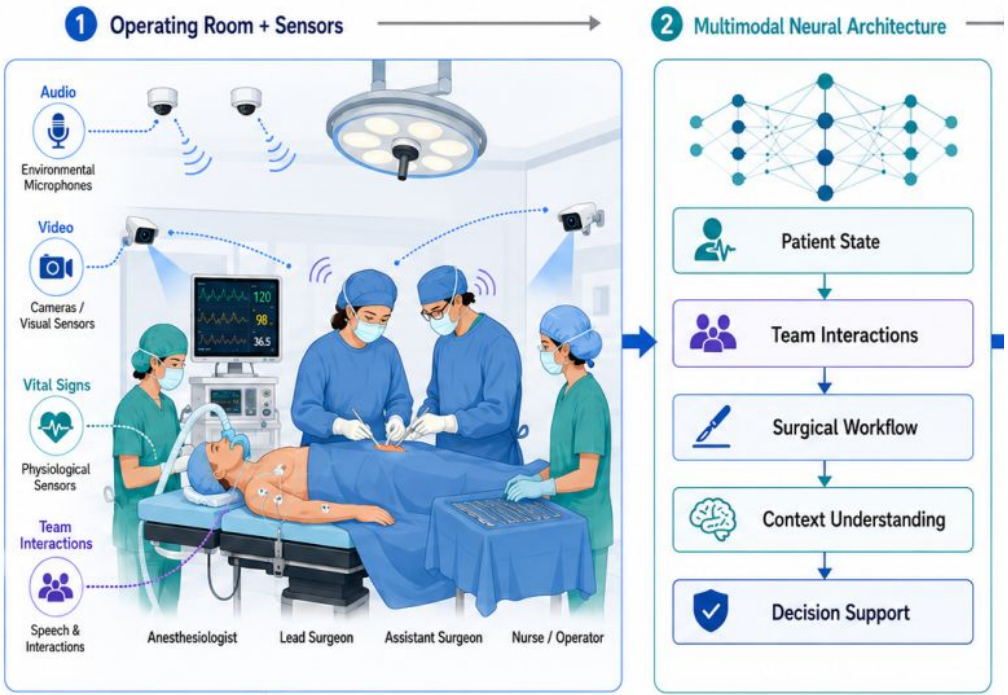
But difficult to automatically analyze them due to:

1. Lack of **public datasets** and **annotations** capturing interactions in the OR (Sec. 2)
2. Lack of **proper neural architectures** modeling team behavior (Sec. 3)
3. Lack of systems able to provide **feedback** (Sec.4)

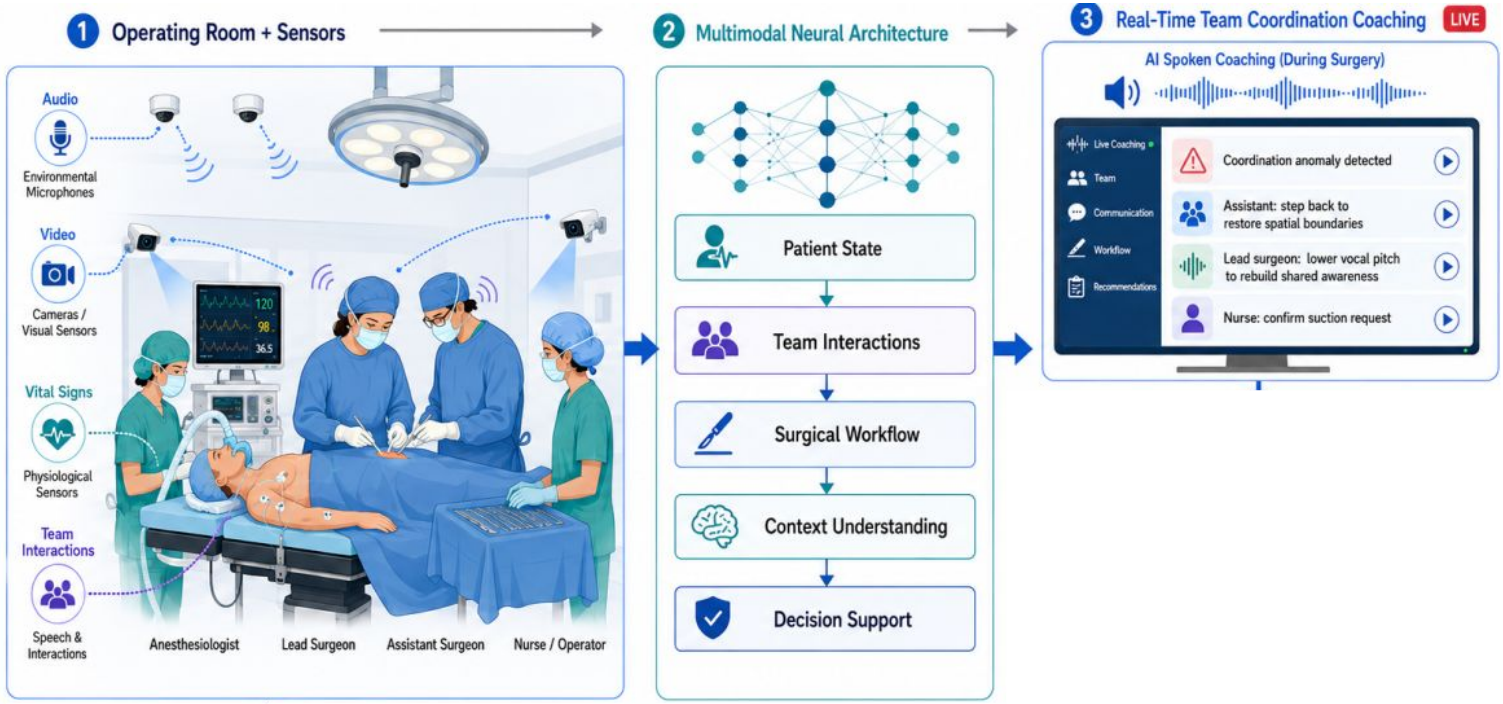
1. Operating Rooms 2.0 - Design



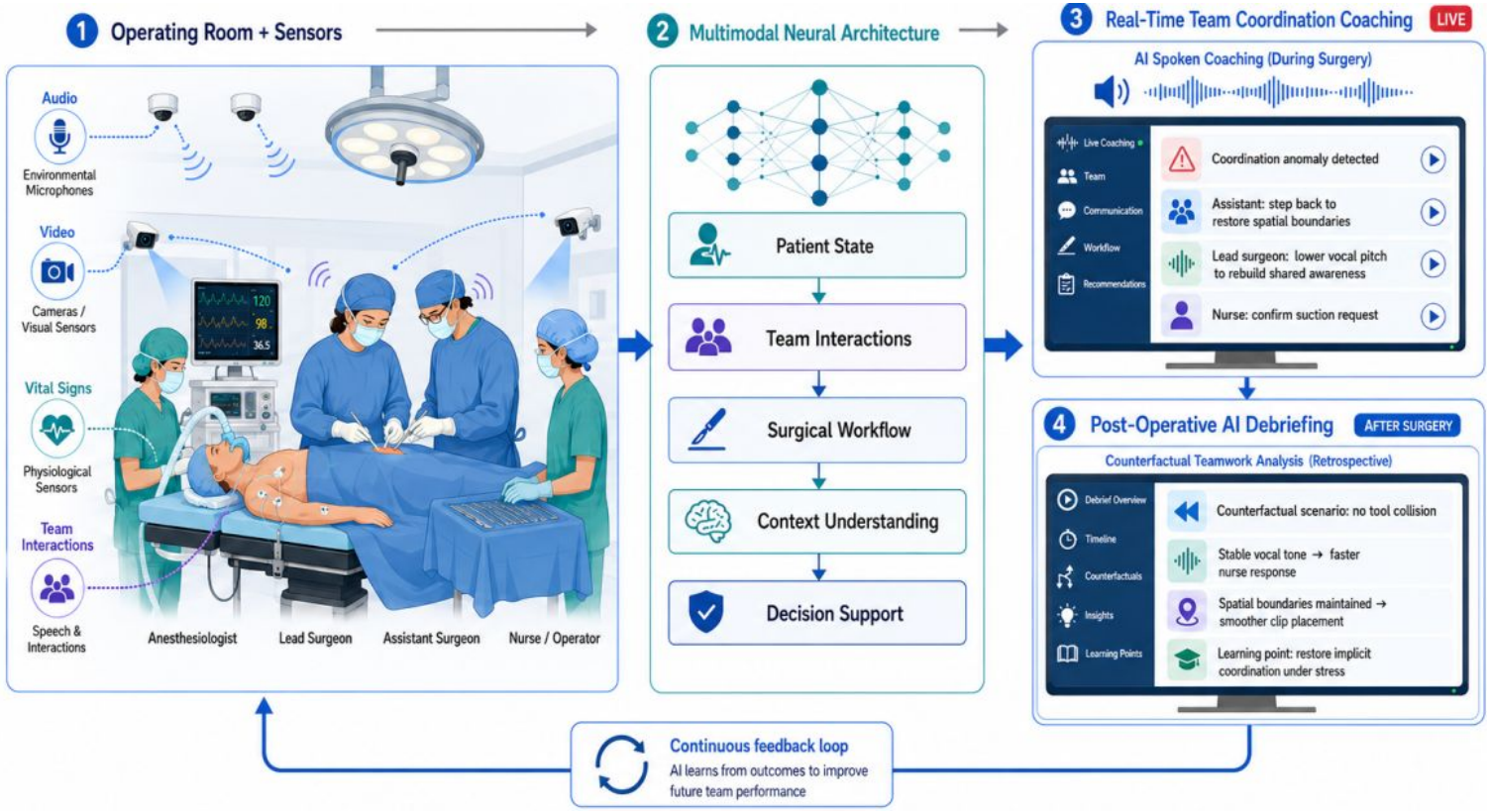
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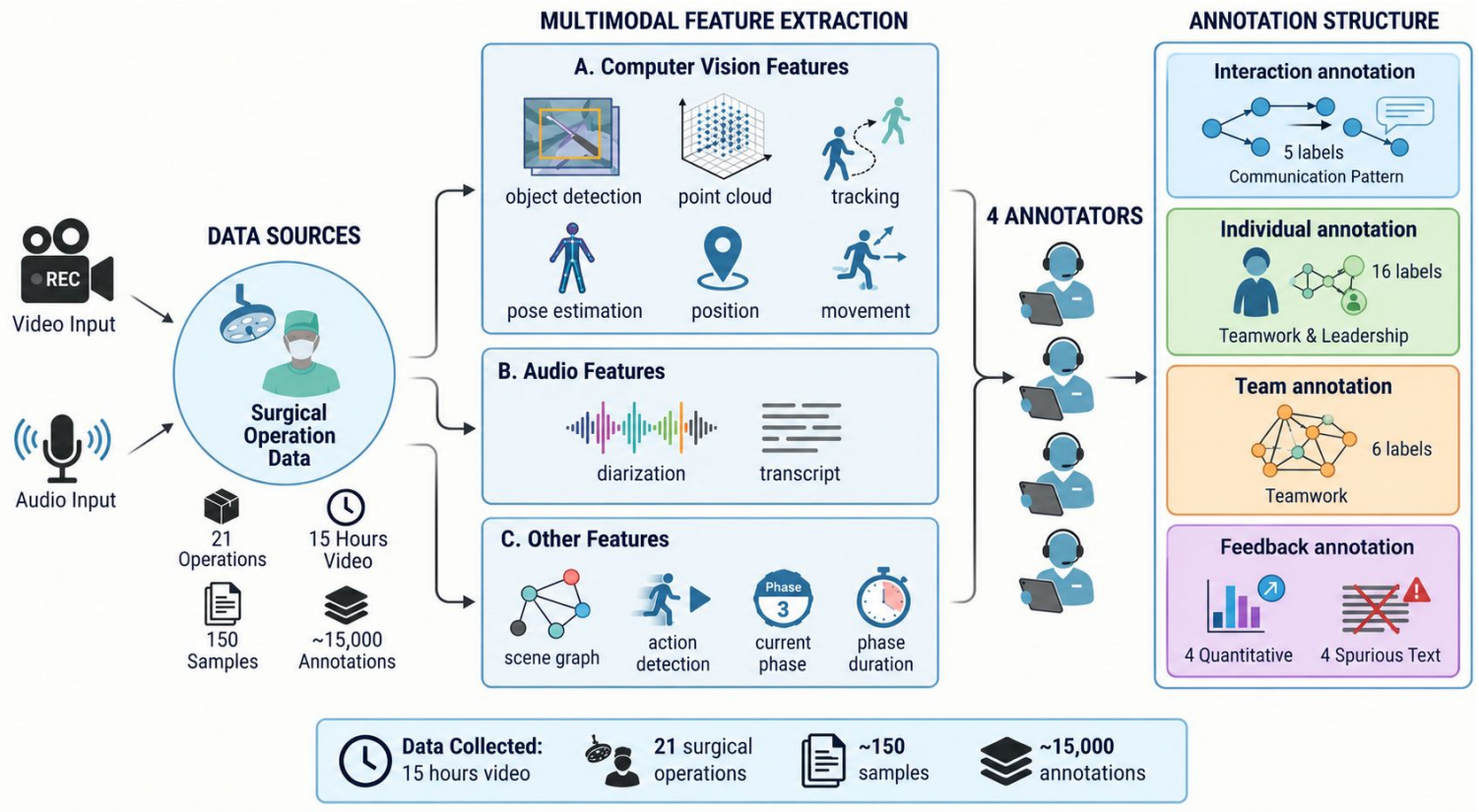
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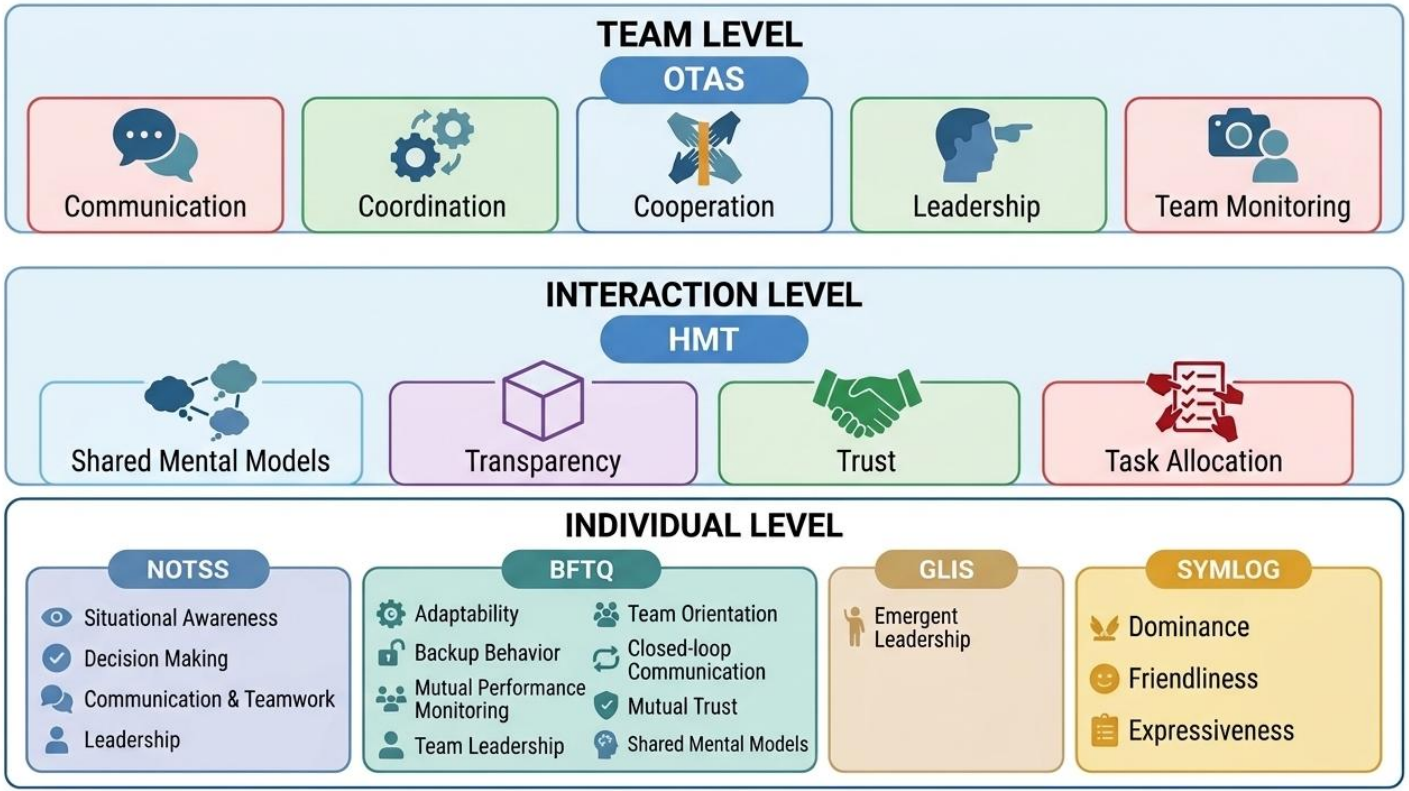
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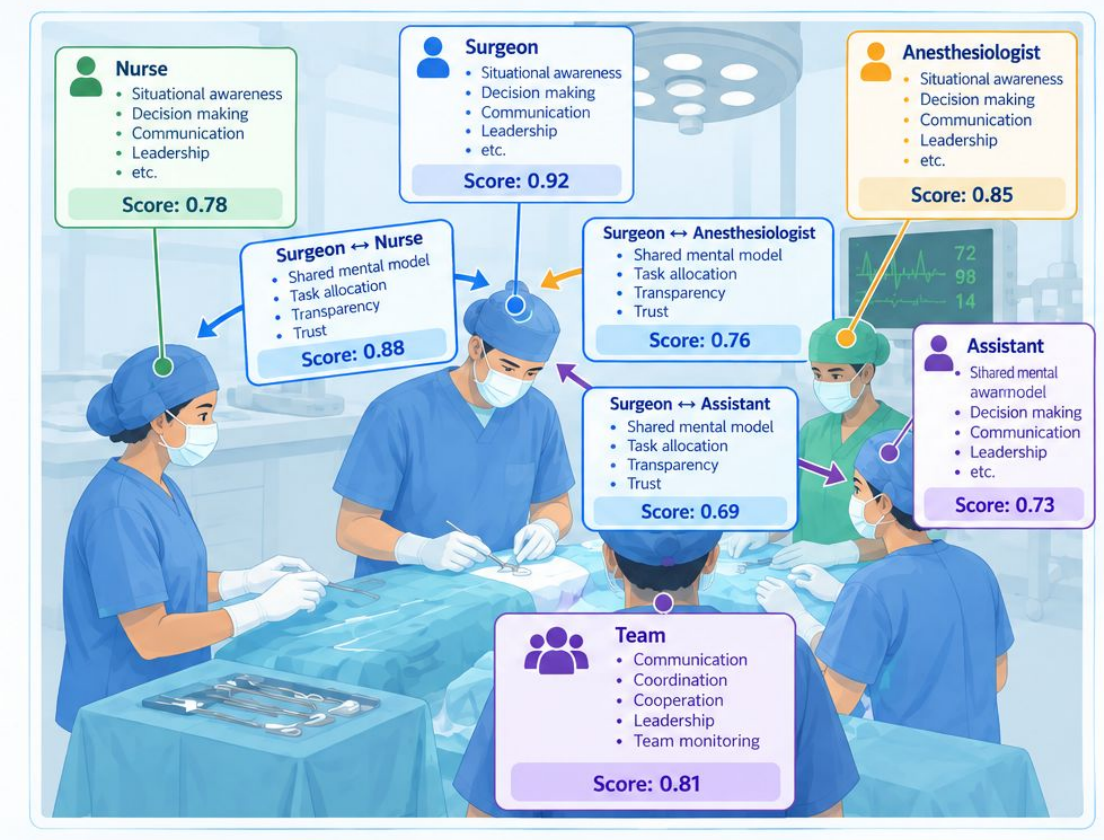
2. Dataset - Data collection



2. Datasets - Data annotation



2. Datasets - Data annotation example



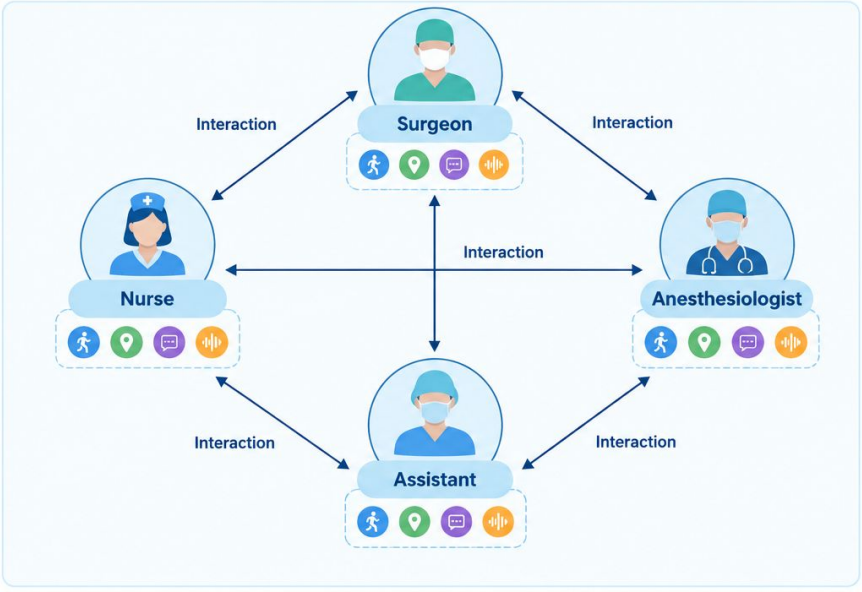
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3. Modeling teams - Static graphs

Static Graph

A snapshot of the surgical team at a given time.



Node features
(multimodal)

Movement
body pose, gestures, movements

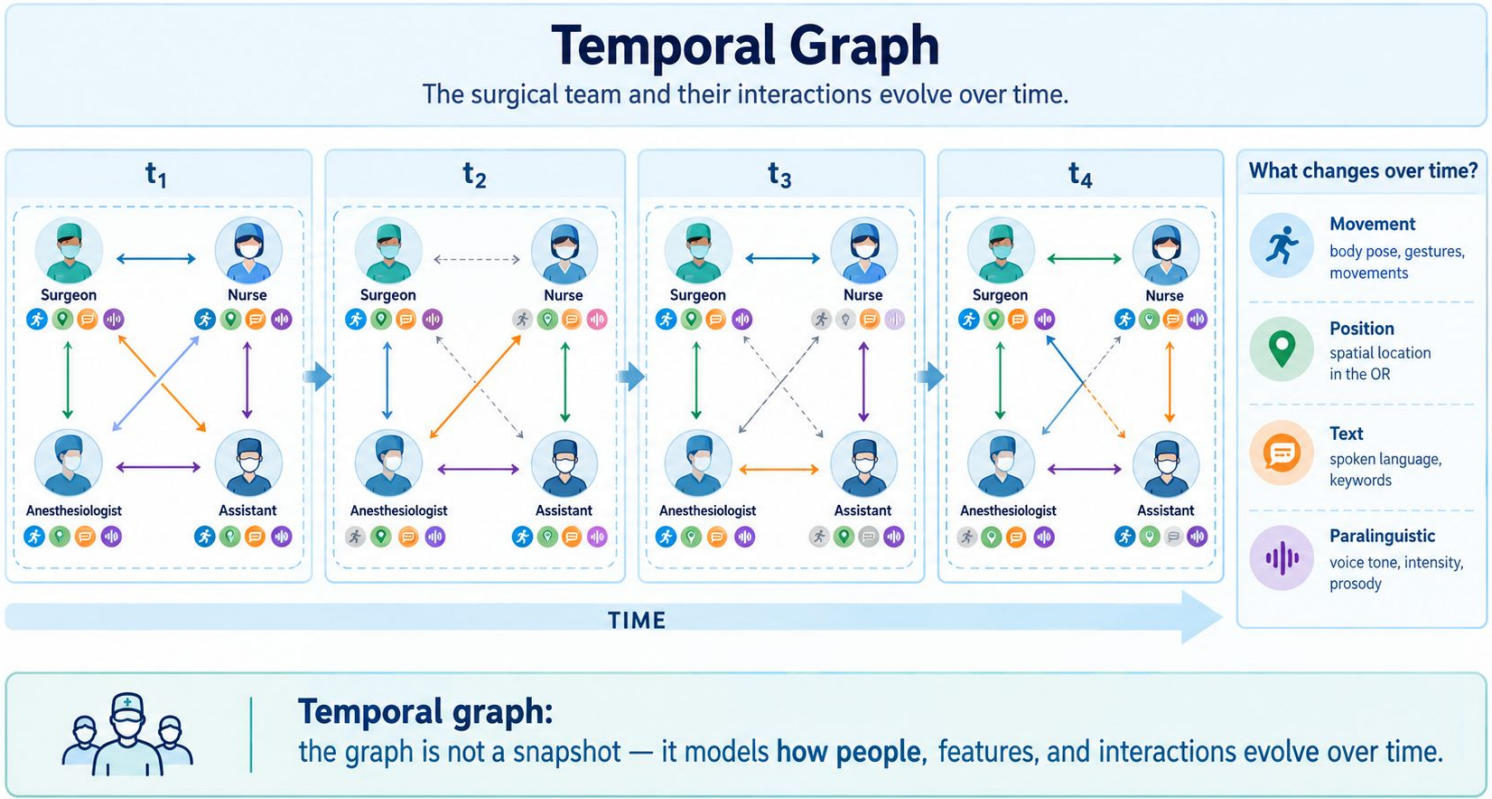
Position
spatial location in the OR

Text
spoken language, keywords

Paralinguistic
voice tone, intensity, prosody

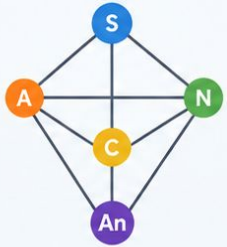
Static graph: nodes represent people with multimodal features; edges represent interactions between team members.

3. Modeling teams - Temporal graphs



3. Modeling teams - Static vs Temporal graphs

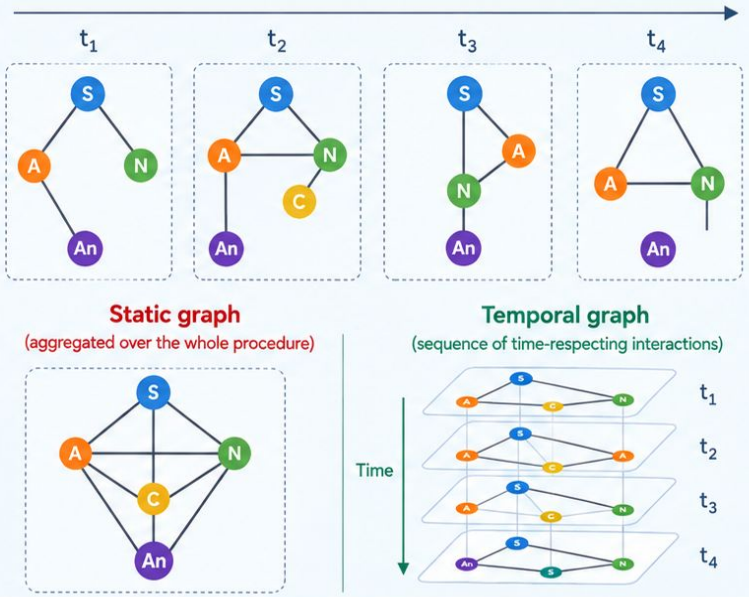
Static Graph



- Treats interactions as persistent
- Loses event order
- Misses transient collaborations

Temporal Graph

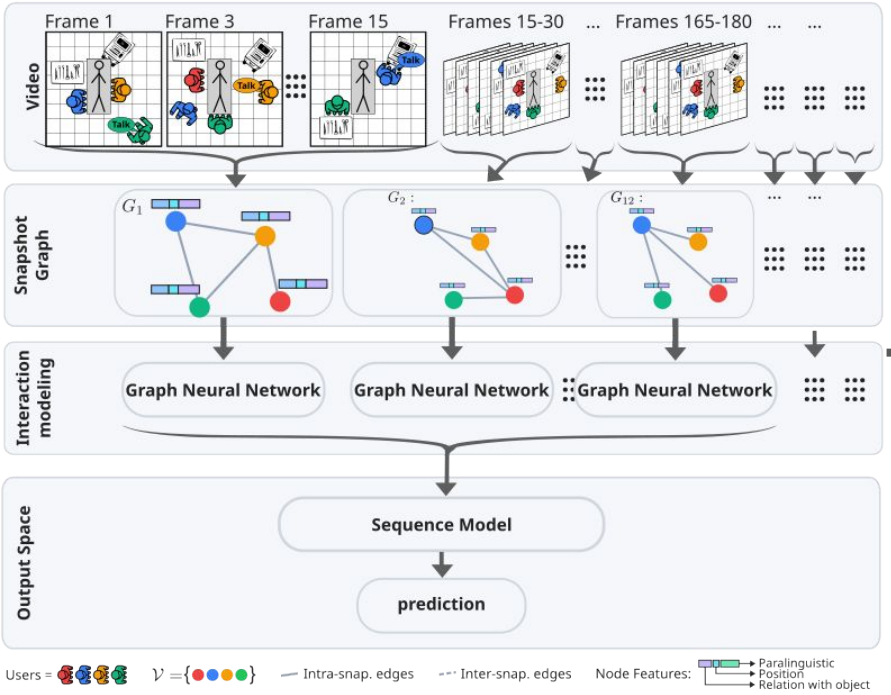
- Captures when interactions occur
- Preserves interaction sequence
- Reveals changing team dynamics

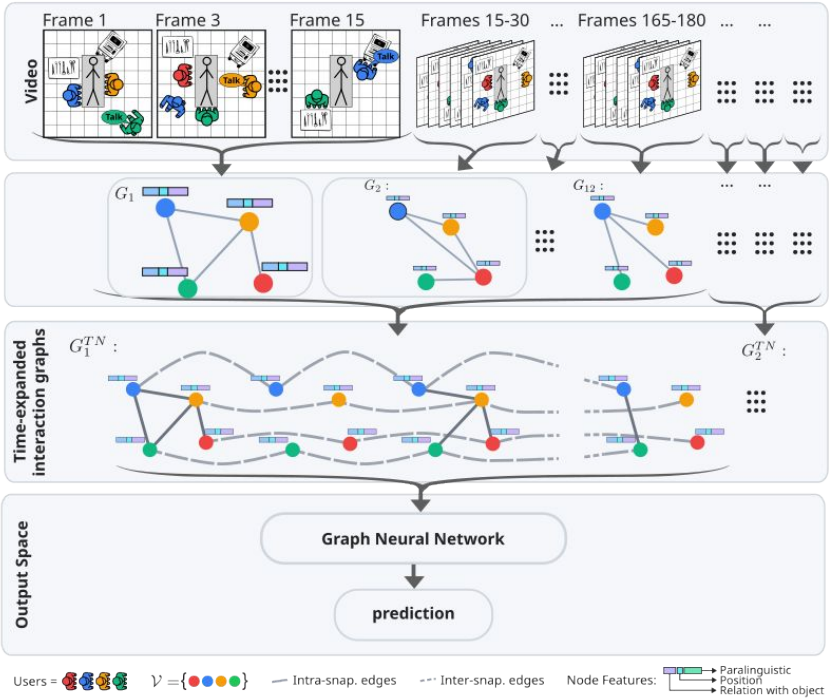


Challenge

Small teams • Limited temporal data

3. From Temporal To Time-Expanded GNN





3. Time-Expanded GNN performance

Paradigm	Model	Team		Individual				Interaction
		OTAS	Phase duration	NOTSS	BFTQ	SYMLOG	GLIS	HMT
SNN	MLP	59.8 ± 2.2	60.5 ± 1.2	51.5 ± 1.0	42.5 ± 1.1	48.5 ± 0.9	64.5 ± 0.8	41.5 ± 1.2
	RF	60.2 ± 0.8	61.8 ± 1.8	52.1 ± 1.2	38.5 ± 0.7	42.1 ± 0.6	65.2 ± 0.7	44.1 ± 1.0
TNN	LSTM	62.3 ± 1.6	62.8 ± 0.9	56.2 ± 1.4	52.2 ± 1.3	55.2 ± 0.9	64.8 ± 1.1	48.2 ± 1.1
	GRU	61.8 ± 1.8	63.0 ± 0.8	58.1 ± 1.1	55.5 ± 1.2	57.3 ± 0.8	62.6 ± 1.3	51.2 ± 1.3
	MHA	63.8 ± 2.0	62.4 ± 1.3	60.4 ± 1.2	56.2 ± 1.0	61.3 ± 0.8	64.9 ± 1.0	53.0 ± 1.2
RENN	GCN	61.4 ± 1.5	64.6 ± 0.6	58.2 ± 1.5	57.5 ± 1.1	59.8 ± 0.9	63.7 ± 1.2	56.2 ± 1.2
	GAT	62.1 ± 1.8	65.2 ± 0.8	59.3 ± 1.0	58.5 ± 1.2	59.6 ± 1.0	65.4 ± 1.2	58.0 ± 1.2
	GIN	65.2 ± 1.5	64.0 ± 1.2	61.2 ± 1.1	59.0 ± 1.3	60.4 ± 1.1	66.8 ± 1.0	58.3 ± 1.1
Space-and-Time TRENN	GCN+MHA	64.0 ± 1.5	67.3 ± 0.6	66.6 ± 1.1	62.8 ± 1.0	64.1 ± 0.8	62.8 ± 1.2	65.4 ± 1.6
	GAT+MHA	67.2 ± 1.7	67.1 ± 0.7	65.6 ± 1.1	65.8 ± 1.2	66.1 ± 1.1	62.0 ± 0.8	65.1 ± 1.2
	GIN+MHA	63.2 ± 1.5	65.2 ± 0.6	66.2 ± 1.0	65.2 ± 1.1	66.0 ± 1.2	64.2 ± 1.0	65.2 ± 1.3
Time-and-Space TRENN	MHA+GCN	65.2 ± 1.3	64.8 ± 0.8	64.0 ± 1.2	65.2 ± 1.8	63.8 ± 1.0	65.2 ± 0.9	69.4 ± 1.1
	MHA+GAT	65.8 ± 1.5	64.8 ± 1.0	64.1 ± 1.0	65.1 ± 1.6	64.0 ± 1.2	67.6 ± 0.9	66.8 ± 1.1
	MHA+GIN	64.2 ± 1.2	65.6 ± 0.9	65.0 ± 1.2	64.1 ± 1.4	63.2 ± 1.1	68.4 ± 1.0	67.9 ± 1.0
TE-RENN	TE-GCN	72.4 ± 1.7	<u>70.1 ± 1.0</u>	71.0 ± 1.3	76.1 ± 1.4	70.2 ± 1.1	75.4 ± 1.0	73.9 ± 1.2
	TE-GAT	<u>73.1 ± 1.7</u>	69.8 ± 0.8	71.7 ± 1.2	75.5 ± 1.1	71.5 ± 1.1	74.8 ± 1.0	**73.9 ± 0.8**
	TE-GIN	**74.2 ± 1.7**	**72.5 ± 1.0**	**72.5 ± 1.0**	76.7 ± 1.1	73.9 ± 1.0	**76.2 ± 1.2**	73.5 ± 0.8

Table 6: Combined results measuring F1-macro according to a LOGO protocol, with experiments runned over 10 different random seeds. Tasks are organized into three levels: Team (OTAS, Phase duration), Individual (NOTSS, BFTQ, SYMLOG, GLIS), and Interaction (HMT). Bold indicates best, underline indicates second best.

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4. Feedback - Explainability p.o.v.

FACTUAL EXPLAINABILITY

Why did the AI give this prediction?



AI PREDICTION:
HIGH RISK
of conflict in the surgical team

WHY?

Higher-than-expected number of surgeons on standby	+0.42
Recent tense communication in the last 24 hours	+0.31
Surgery duration expected to be longer than usual	+0.21

The model flagged this risk because of these observable **team conditions and behaviors** (e.g., more people, recent tension, longer case).

VS

COUNTERFACTUAL EXPLAINABILITY

What could change the prediction?



AI PREDICTION:
LOW RISK
of conflict in the surgical team

WHAT COULD CHANGE THE PREDICTION?

If the following changes:

Reduce number of surgeons on standby (e.g., from 5 to 2) →	RISK DECREASES **LOW**
Resolve recent tension (e.g., through structured brief) →	RISK DECREASES **LOW**
Shorten expected surgery duration (e.g., optimize schedule) →	RISK DECREASES **LOW**

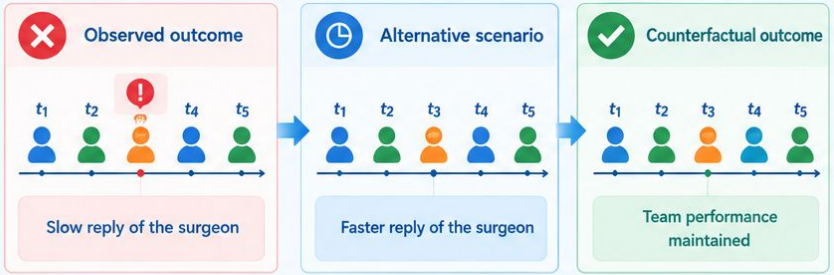
If these **team behaviors** change, the likelihood of conflict drops significantly.

4. Feedback - Annotation

What are counterfactual annotations?

Identify a plausible alternative team outcome
for an observed interaction failure.

Example



Annotation dimensions

- Criticality
- Plausibility
- Confidence
- Minimality
- Expected enhancement

Annotation protocol

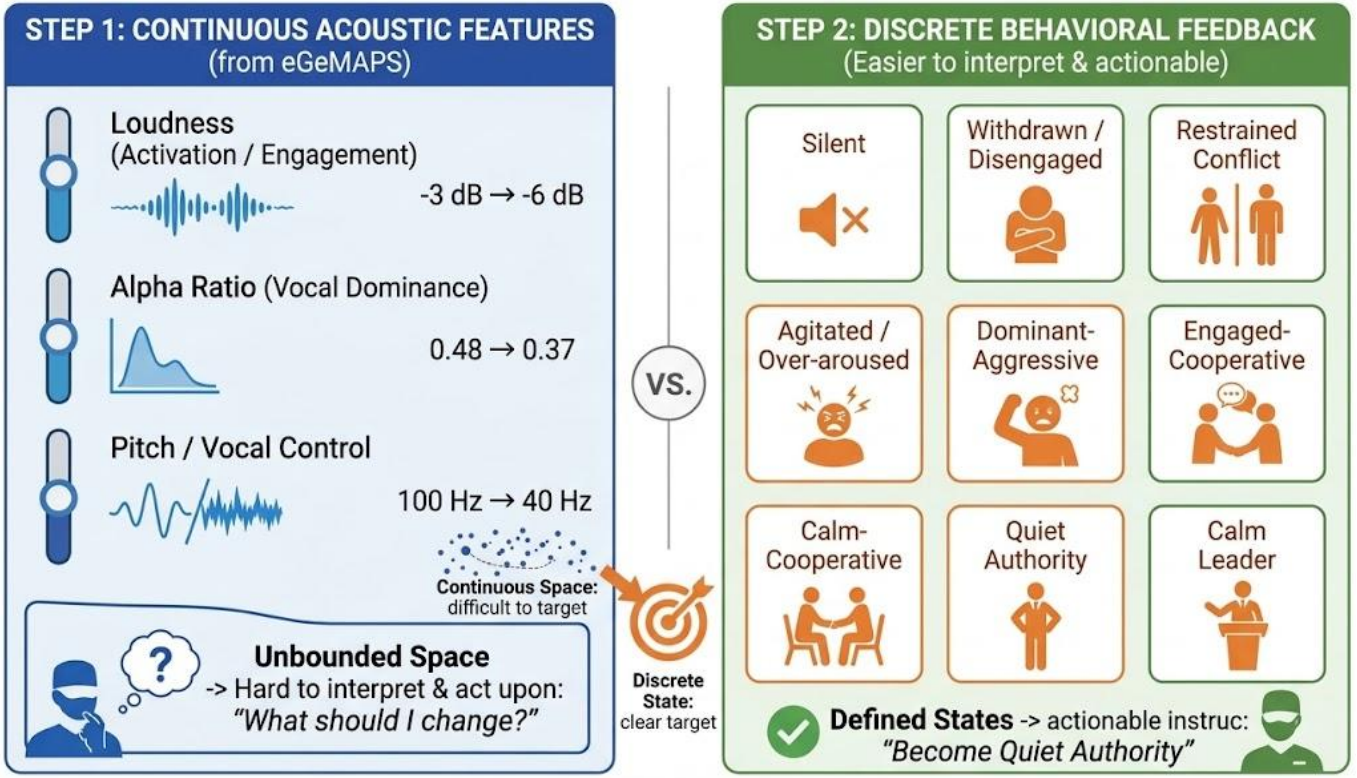
Structured fields

- Critical category (OTAS)
- Time interval
- Type
- Responsible person(s)

Open-ended questions

- What happened?
- How to adjust?
- What did it cause?
- Why avoid it?

4. Feedback - Getting interpretable



Counterfactual algorithm

Algorithm 1 Actionable Counterfactual Generation

Require: Input prefix $\mathcal{G}_{1:t}$, target outcome y^{target}, encoder f_θ, repository $\mathcal{Z}$
Ensure: Actionable counterfactual explanation e

Offline: Build repository $\mathcal{Z}$
for each training sequence $\mathcal{G}^i_{1:T_i}$ **do**
  $z^i \leftarrow f_\theta(\mathcal{G}^i_{1:T_i})$
end for
Online: Encoding
$\mathbf{z} \leftarrow f_\theta(\mathcal{G}_{1:t})$
Retrieval of the most similar sample
$(\mathcal{G}^*_{1:t}, y^{\text{retrieved}}) \leftarrow \textsc{Retrieve}(\mathbf{z}, \mathcal{Z}, y^{\text{target}})$
Factual explanation of target samples
$\mathcal{E} \leftarrow \textsc{Explain}(\mathcal{G}^*_{1:t}, f_\theta)$
Selection
for each explanation $e \in \mathcal{E}$ (ranked by importance) **do**
  if $\textsc{Match}(e, \mathcal{G}_{1:t}) =$ **false then**
    return e
  end if
end for
return $\emptyset$

4. Feedback - Evaluation

1) Adding Communication



Situation. The nurse asks a question, but the leader is distracted and does not respond, leading to silence and low communication.

Counterfactual. Add edges

Interpretation. Reduce disengagement by increasing leader responsiveness.

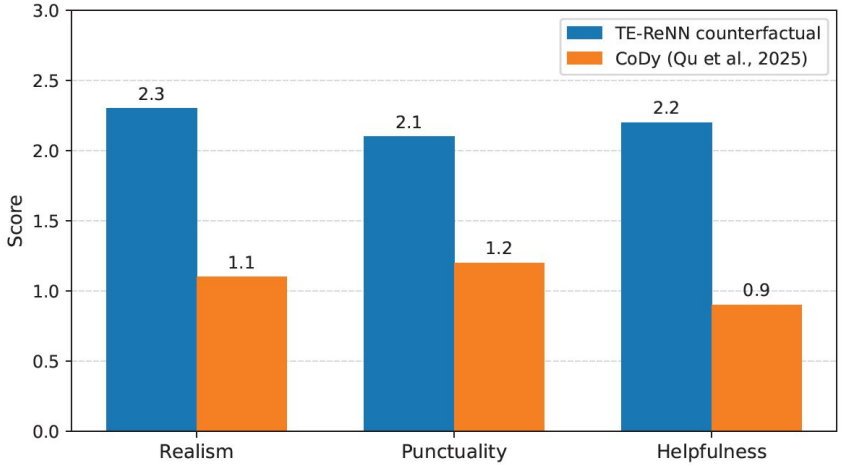
4) Aggressive Communication



Situation. A rude response from the leader to a nurse's question reduces team cooperation.

Counterfactual. Increase "Control" (low → high).

Interpretation. Adopt a "Calm Leader" style instead of "Dominant–Aggressive".



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+ Are there more suitable architecture for modeling small team interactions?
+ Is it possible to realize continuous optimize of counterfactual explanation?
+ Is it possible to forecast how does team behavior evolve in time?
+ Is it possible to validate this explanation in a trainees scenario in a simulated OR?

Thank you for the attention!

Boosting Team Modeling through Tempo–Relational Representation Learning, Cognitive Communication, 2025

Towards Actionable Surgical Team Dynamics: from Teamwork to Counterfactual Annotations, Human-AI Complementarity and Alignment, 2026

Actionable Real-Time Modeling of Surgical Team Dynamics via Time-Expanded Interaction Graphs , Human-Hybrid Artificial Intelligence, 2026

From Multimodal Observation to Interpretable Suggestions: Counterfactual Time-Expanded Relational Modeling of Surgical Teams, ACM Multimedia, 2026

Appendix - Behavioral classes

Activation	Control	Dominance	Behavioral Class
Low	Low	Low	Withdrawn / Disengaged
Low	Low	High	Restrained Conflict
Low	High	Low	Calm–Cooperative
Low	High	High	Quiet Authority
High	Low	Low	Agitated / Over-aroused
High	Low	High	Dominant–Aggressive
High	High	Low	Engaged–Cooperative
High	High	High	Calm Leader

Table 1: Behavioral classes defined by combining three paralinguistic features extracted from the eGeMAPS set.